Show all work and calculations

1) Consider a 100-MHz uniform plane wave propagating in a lossy non-magnetic medium with $\gamma = 0.2 + j3.4$ (1/m). The electric field is given by

$$\vec{E} = \left(j5\,\vec{a}_x + 12\,\vec{a}_y\right) e^{-\gamma z} \quad \text{(V/m)}$$

a) Find the polarization of the wave (linear, circular, etc.)
b) Compute the impedance of the medium.
c) Find the corresponding magnetic field expression.

2) A plane wave propagating in a lossless non-magnetic dielectric material with $\varepsilon_r = 2.25$ is incident upon the planar surface of another lossless non-magnetic dielectric material with $\varepsilon_r = 4$ occupying the half-space $x \geq 0$. The electric field of the incident wave is given by

$$\vec{E} = 20\,\vec{a}_y\, e^{-j(3x+4z)} \quad \text{(V/m)}$$

a) Produce a sketch of the problem.
b) What is the polarization of the wave (TE/TM)?
c) Find the angle of incidence.
d) Find the frequency of the wave.
e) Write the phasor expression of the reflected <u>electric</u> field.
f) Write the instantaneous expression of the transmitted <u>magnetic</u> field.

3) It is required to design an anti-reflection coating at 0.9 GHz for a ground penetrating radar application. Assuming the EM waves originate in air and hit the soil (ε_r=6) normally, find a suitable and practical coating thickness (in cm) and the dielectric constant of the coating material.